

# Category-specific benchmarking of Scope 3 emissions for corporate clusters

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## ABSTRACT

Although the majority of corporate emissions is assigned to the upstream and downstream value chain (Scope 3) rather than to direct emissions (Scope 1) and emissions from energy purchases (Scope 2), Scope 3 data are often insufficient, and their collection is complex due to many reference variables. Therefore, Scope 3 reporting by more than 4000 companies worldwide in 2022 is examined using data from the CDP (formerly Carbon Disclosure Project). All fifteen categories of this Scope as defined by the Greenhouse Gas Protocol are considered. To enhance the benchmarking results, corporate clusters based on size and industry are built after analyzing which factors significantly influence Scope 3.

Based on the results, a benchmarking for the emissions per yearly turnover is carried out and examined on five case studies. As the manufacturing industry comprises various subindustries ranging from plastics to automotive, the cluster consisting big enterprises is further subdivided. Based on two case studies, it is possible to demonstrate that emission hotspots may be situated in diverse categories although companies belong to the same cluster. Companies can use the results for an initial estimation of category-specific Scope 3 emissions, e.g., regarding their sustainability reporting. The novelty of this work is a category-specific benchmarking of Scope 3 emissions considering corporate clusters based on companies worldwide and considering industry and size.

## 1. Introduction

According to the latest Synthesis Report by the Intergovernmental Panel on Climate Change, taking action to limit global warming to 1.5 °C and mitigate its impacts on the earth's ecosystems is more crucial than ever before [1]. This is particularly relevant for companies as their production and energy demand contribute significantly to global greenhouse gas (GHG) emissions [2]. In 2021, 58 % of European citizens considered business and industry to be responsible for addressing climate change, which represents a 7 % increase from 2019 [3]. This illustrates that not only the political pressure, such as through national climate regulations, on companies to address climate change is rising but also the social pressure.

Since indirect GHG emissions from a company's supply chain, referred to as Scope 3 emissions by the Greenhouse Gas Protocol (GHGP) [4], outweigh direct emissions (Scope 1) and those from purchased energy (Scope 2), attention is paid to this part of the corporate carbon footprint [5]. However, Scope 3 data is often insufficient [6]. Information on GHG emissions is frequently lost along the supply chain [7], leading to a dependence on literature values. Furthermore, Scope 3 is more extensive than Scope 1 and 2, resulting in a complex and

time-consuming data collection. Nevertheless, an accurate Scope 3 balance can facilitate interaction between different companies along their value chain to address climate protection issues.

Due to these reasons, a method is needed that can be used to fully estimate the GHG emissions of Scope 3. This method must be simple to involve as little work as possible, and at the same time offer the possibility of estimating Scope 3 as accurately as possible. Several approaches have already investigated such methods. For example, Klaaßen and Stoll develop a harmonizing framework for assessing Scope 3 emissions focusing on the tech sector [8]. Schmidt et al. propose an economic input-output analysis [7], while Montoya-Torres et al. develop a conceptual framework using a case study research approach [9] for the assessment of Scope 3. Using CDP (formerly Carbon Disclosure Project) data from 2013, Blanco et al. develop a benchmarking method to assess the Scope 3 reporting of companies in the United States [10]. Rizet et al. carry out a benchmarking approach referring to value chains and Scope 3 emissions of the transport in the food and textile industries in different countries [11]. Acquaye et al. use a multi-regional input-output analysis for the development of supply chain carbon maps with the aim of assessing Scope 3 emissions and identifying emissions hotspots [12]. Across these approaches, especially the benchmarking stands out. With this method the average percentage shares of individual areas in Scope 3

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**Nomenclature****Abbreviations**

|                   |                                          |
|-------------------|------------------------------------------|
| BE                | big enterprise                           |
| CDP               | Carbon Disclosure Project (formerly)     |
| CO <sub>2</sub> e | carbon dioxide equivalents               |
| GHG               | greenhouse gas                           |
| GHGP              | Greenhouse Gas Protocol                  |
| SME               | micro, small and medium-sized enterprise |

**Symbols**

|           |                                                           |
|-----------|-----------------------------------------------------------|
| $C_i$     | number of options within a characteristic $i$             |
| $C_R$     | characteristic factor “relevant”                          |
| $C_{R,c}$ | characteristic factor “relevant and calculated”           |
| $E$       | margin of error                                           |
| $G$       | geographic location                                       |
| $I$       | industry                                                  |
| $N$       | total amount of responses                                 |
| $N_R$     | total amount of responses regarding companies identifying |

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
|              | a category as relevant                                                        |
| $n$          | minimum sample size                                                           |
| $n_{corr}$   | corrected minimum sample size                                                 |
| $n_{corr,R}$ | corrected minimum sample size regarding the characteristic “relevant”         |
| $n_R$        | minimum sample size regarding the characteristic “relevant”                   |
| $p_R$        | share of the characteristic “relevant” regarding all responses                |
| $p_{R,c}$    | share of the characteristic “relevant and calculated” regarding all responses |
| $S$          | company size                                                                  |
| $z$          | width of the confidence interval                                              |

**Units**

|     |                |
|-----|----------------|
| EUR | euro           |
| t   | metric ton     |
| °C  | degree Celsius |

can be determined based on companies’ information on their GHG emissions. One reason for this is the ease of application which eliminates the need for extensive prior analysis of a company’s value chain. Moreover, benchmarking does not require a broad basis of primary data. In addition, this method can be easily adapted to changing circumstances.

Previous studies that have examined the assessment of Scope 3 emissions have done so against the background of specific sectors, the location of the company as well as its size. In this regard, Blanco et al. conclude that Scope 3 and corresponding reporting varies greatly depending on the industry within the United States [10]. Bolay et al. compare sectoral GHG emissions based on mitigation targets [13]. Although they focus on Scope 1 and Scope 2 the sector seems to influence corporate emissions. Córdova et al. investigate the relationship between different corporate characteristics of South American companies and their emissions reporting and development [14]. The study shows that company size and sector in particular have a positive influence, while the opposite can be the case in relation to the country [14]. Therefore, these three factors – industry, company size, and geographic location – are identified as potential influence for Scope 3 emissions.

However, a detailed investigation whether these three factors actually influence Scope 3 emissions based on a broad data basis that is not limited to a certain region is still missing. Moreover, until now there are no holistic corporate benchmarks that consider the influencing factors as well as different Scope 3 areas, such as the purchase of goods and services. These areas are defined by the GHGP and consist of fifteen distinct categories [4]. The results in the form of percentage shares offer the possibility of fully estimating Scope 3, even if only individual categories are known. Furthermore, when examining which categories are relevant for which companies, such benchmarks could also be used to identify emissions hotspots just like Acquaye et al. [12] did based on another approach. Hence, this study aims to develop such benchmarks that are based on a broad database. For this, CDP data from the year 2022 is evaluated. Such a broad approach is necessary to fully address and compare different value chains. As it is initially assumed that the accuracy of the benchmarks can be increased by considering factors that influence Scope 3 emissions, corporate clusters are built that take these into account. For determining such factors, at first, the three factors identified based on the relevant literature – industry, company size, and geographic location – are evaluated regarding their impact on the extent of emissions. This could lead to a better understanding of the global emissions landscape and help to develop more effective and more widely applicable reduction strategies as well as reporting standards. Moreover,

the results can be used for an initial estimate of the GHG emissions’ magnitude of a particular Scope 3 category for a company in the identified cluster.

This study is structured as follows: Section 2 provides an overview on the current state-of-the-art for assessing Scope 3 emissions based on a systematic literature review. Section 3 describes the considered materials and methods. In section 4, the results are presented, which are then discussed in section 5 and summarized in section 6.

## 2. Evaluating Scope 3 emissions

In this section a literature review is given regarding the assessment and evaluation of Scope 3 and corresponding emissions. This review is carried out based on a keyword search via Google Scholar and taking into account the first 20 results for each keyword. With regard to the research topic, the keywords included “Scope 3 assessment”, “Scope 3 benchmarking”, “corporate clustering”, and “corporate cluster”. Next to the findings of this search, also further scientific literature is considered as already known relevant literature to complete a comprehensive review.

**Table 1**

Overview of the scientific literature with regard to the criteria scope of investigation and assessment of GHG emissions.

| study | scope of investigation |               |            | assessment of GHG emissions |        |
|-------|------------------------|---------------|------------|-----------------------------|--------|
|       | cross-industry         | cross-country | clustering | benchmarking                | others |
| [7]   | ●                      |               |            |                             | ●      |
| [8]   |                        | ●             | ●          |                             | ●      |
| [9]   |                        | ●             |            | ●                           | ●      |
| [10]  | ●                      |               | ●          | ●                           | ●      |
| [11]  | ●                      | ●             |            | ●                           |        |
| [12]  |                        | ●             |            | ●                           |        |
| [13]  | ●                      | ●             | ●          | ●                           |        |
| [14]  | ●                      | ●             |            |                             | ●      |
| [15]  |                        | ●             |            |                             | ●      |
| [16]  | ●                      | ●             |            |                             | ●      |
| [17]  |                        | ●             | ●          |                             | ●      |
| [18]  |                        | ●             |            |                             | ●      |
| [19]  |                        | ●             |            | ●                           |        |
| [20]  |                        | ●             |            | ●                           |        |
| [21]  |                        |               |            | ●                           |        |
| [22]  |                        |               |            | ●                           |        |
| [23]  |                        |               |            | ●                           |        |
| [24]  | ●                      |               | ●          |                             | ●      |

Table 1 provides an overview of the studies with regard to the scope of their investigation, which is subdivided into cross-industry, cross-country, and clustering of companies based on certain characteristics. In this regard, cross-industry refers to studies that focus on more than one industry. Analogously, cross-country refers to those studies that consider multiple countries. When a study clusters companies based on different characteristics this is shown in the column clustering under the scope of investigation. The consideration of clusters is independent of the method used for clustering. In addition, Table 1 shows which studies have benchmarked greenhouse gas emissions and which studies have chosen a different method to determine them. The differentiation is made only between benchmarking and other methods as benchmarking is the most frequently used method. Other methods for assessing GHG emissions are, for example, economic input-output-analysis [7] as well as Environmental Input-Output Life-Cycle Assessment models [10], Chi-square tests [15], or newly developed approaches [9].

Table 1 shows that only one of the studies takes a cross-sector and cross-country view into account, then assigns companies to individual groups and carries out a specific benchmarking for these groups. Nevertheless, this study focuses on Scope 1 and Scope 2 emissions [13], whereas the focus of the following analyses explicitly refers to Scope 3 emissions. Regarding other studies that have carried out a benchmarking, this was in most cases not done for Scope 3 emissions, but referred to, for example, cities [21]. Other studies did not benchmark Scope 3 emissions themselves but described this as a possibility that should be given further consideration [16]. Regarding the clustering of companies, it can be noted that the considered studies take individual industries as well as a cross-industry approach into account equally. Bolay et al. [13] as well as Blanco et al. [10] focus on the industry as factor for distinguishing different companies from each other.

The results of this literature review underline the need for a standardized method to estimate Scope 3 with as little effort as possible for companies. However, to ensure representative results, some studies consider company clusters based on certain characteristics. A comprehensive study based on companies worldwide and various sectors to determine which corporate characteristics actually have an influence on Scope 3 is currently missing and should, thus, be carried out.

### 3. Materials and methods

To address the identified research gap, company-specific data regarding Scope 3 reporting and corresponding GHG emissions of more than 4000 companies are analyzed. In this regard, Scope 3 categories and corresponding emissions are considered in more detail. Afterwards the methodological basis of the cluster analysis and benchmarking is described.

#### 3.1. Data basis

For analyzing Scope 3 reporting and corresponding emissions on a category-specific basis, climate data of the CDP for the year 2022 are evaluated. This section relies on the definition set forth by the GHGP which is described in the following. Depending on the point in the value chain at which the emissions are generated, Scope 3 can be segregated into two sections: upstream and downstream emissions. Each of these sections is further subdivided into eight and seven categories, respectively. A list of the individual categories with the respective allocation to upstream and downstream activities can be found in Table 2. The Technical Guidance for Calculating Scope 3 Emissions of the GHGP provides a brief overview of the emission sources per category [25].

The answers of more than 8300 companies of different sectors and from all over the world to the specific 2022 CDP climate questionnaire are considered with a special focus on Scope 3 reporting. In this regard, the questionnaire first asks companies to identify whether a Scope 3 category (Table 2) is relevant or not considering their individual corporate activities. The companies could select one of the following

**Table 2**

Scope 3 categories differentiated into upstream and downstream emissions according to the GHGP [25].

| upstream activities                                                       | downstream activities                         |
|---------------------------------------------------------------------------|-----------------------------------------------|
| 1. Purchased Goods and Services                                           | 9. Downstream Transportation and Distribution |
| 2. Capital Goods                                                          | 10. Processing of Sold Products               |
| 3. Fuel- and Energy-Related Activities Not Included in Scope 1 or Scope 2 | 11. Use of Sold Products                      |
| 4. Upstream Transportation and Distribution                               | 12. End-of-Life Treatment of Sold Products    |
| 5. Waste Generated in Operations                                          | 13. Downstream Leased Assets                  |
| 6. Business Travel                                                        | 14. Franchises                                |
| 7. Employee Commuting                                                     | 15. Investments                               |
| 8. Upstream Leased Assets                                                 |                                               |

answer options: *Not evaluated*, *Not relevant*, *Relevant*, *not calculated*, *Relevant*, *calculated*, and *Question not applicable*. Accordingly, each company decides which categories are relevant on its own. Second, if a category was identified as relevant companies can provide further information on GHG emissions per category in metric tons (t) of carbon dioxide equivalents (CO<sub>2</sub>e). This is regardless of whether this category is calculated or not. Moreover, along with this, additional comments can be provided if necessary. For example, such a comment could refer to reasons why only some aspects of a category were considered within the calculation. Nevertheless, there is no consistent standard specified by CDP for the accounting of GHG emissions. This results in companies individually choosing to use for example the GHGP or a sector-specific standard instead. Furthermore, general information about each company is considered in this study as basis for the analysis. This information comprises a company's primary industry, the location of its headquarters, the currency used, and the number of employees as well as the turnover in the reporting year. However, it should be noted that all information provided in the questionnaire is self-reported by the companies and not independently verified by third parties or the CDP.

#### 3.2. Cluster analysis

To compare the responses to the Scope 3 section of the 2022 climate questionnaire among different companies, it is necessary to group the heterogeneous results into homogenous categories. To achieve this, a cluster analysis is deemed appropriate [26]. For this analysis three different characteristics are chosen to be considered: the primary industry, the company size, and the geographical location of the company referring to its headquarters. To analyze whether these factors actually have an influence on the scope of Scope 3 emissions per category, a one-way analysis of variance is carried out. The reason for this is that this method is especially suitable for examining if a nominally scaled factor such as the industry can be assumed to have a significant influence on a dependent metric variable such as GHG emissions [27]. The one-way analysis of variance tests whether a null hypothesis can be accepted or not. The null hypotheses assumes that the variable under investigation has no influence on the emissions per Scope 3 category. For this, the p-value is investigated which is a measure of whether significant differences can be observed or not [28]. It results from the comparison with the significance level, which is used as the threshold value for determining statistical significance [28]. The mathematical principles can be taken from Chandrakantha [28]. In this study, the significance level is set to be 0.05. If the p-value is greater than this significance threshold, the null hypothesis is accepted. If it is less than 0.05, the alternative hypothesis is accepted, which states that there is a significant influence.

A number of methods can be considered to perform a cluster analysis, such as presented by Madhulatha [29]. However, the k-modes method, as presented by Huang [30], is particularly suitable for carrying out a cluster analysis explicitly for nominally scaled data. As it uses those data

directly as such, this method impresses with its efficiency as well as ease of use and provides results that are easy to interpret. To apply the k-modes method, the underlying data must first be prepared in a suitable form [30]. This can be found in Table 3, which shows the various options for industry, company size and geographic location. A detailed explanation of the individual variables is provided in the following. Based on the responses provided by each company, fourteen different industries are identified that need to be differentiated (Table 3).

Regarding the company size, a distinction was made between big enterprises (BE) and micro, small and medium-sized enterprises (SME) (Table 3). In accordance with the European Commission's definition, an enterprise is considered as an SME if it employs less than 250 individuals and has a maximum annual turnover of EUR 50 million [31]. If a company reported its annual turnover in a currency other than euro, these are translated based on the exchange rates as of April 2023. As the questionnaire allowed companies to choose whether to use their 2022 turnover, the number of employees or another unit as reference, in this study the SME definition of the European Commission is modified such that only one of these criteria needs to be met instead of both. Due to some companies choosing not to provide either the number of employees or the turnover, the number of companies that are further considered for this study is reduced by approximately 50 %.

For assigning each company to a specific geographic location, the country in which the company's headquarter is located was used. Subsequently the regions Africa, Asia, Australia, Europe, North America, and South America are identified (Table 3). Companies headquartered in Turkey are assigned to Europe, and those headquartered in Kazakhstan are assigned to Asia.

Afterwards, in the k-modes method the number of clusters are identified by the number of different characteristics. Using equation (1) the number of potential clusters can be calculated based on the number of options for different characteristics  $C$  such as industry, company size, and location.

$$\text{number of combinations} = \prod_{i=1}^n C_i \quad (1)$$

Companies are assigned to the same cluster if their Hamming distance is zero. As the Hamming distance describes the number of differences between individual points [32], it expresses that only companies with the exact same characteristics can be found within a cluster.

### 3.3. Benchmarking

The benchmarking process is conducted on a category-specific basis, including the identified clusters. It is performed in two steps. First, the frequency with which a category is identified as relevant is examined. Second, the quantitative analysis of GHG emissions for each category is performed.

**Table 3**

Characteristics to distinguish the companies from each other and basis for the one-way variance analysis.

| industry                        | size | geographic location |
|---------------------------------|------|---------------------|
| services                        | SME  | Africa              |
| infrastructure                  | BE   | Asia                |
| biotech, health care and pharma |      | Australia           |
| materials                       |      | Europe              |
| manufacturing                   |      | North America       |
| hospitality                     |      | South America       |
| retail                          |      |                     |
| transportation services         |      |                     |
| food, beverage and agriculture  |      |                     |
| power generation                |      |                     |
| apparel                         |      |                     |
| fossil fuels                    |      |                     |
| corporate tags                  |      |                     |
| international bodies            |      |                     |

In the context of assessing the relevance, companies were provided with the following response options: *Not evaluated*, *Not relevant*, *Relevant*, *not calculated*, *Relevant*, *calculated*, *Question not applicable*. Only responses marked as *Relevant*, *calculated* are considered for further analysis. The number of such responses is then compared to the total number of responses ( $N$ ). Subsequently, it is investigated whether this proportion is adequate to provide a representative assessment of the category's relevance for the entire cluster. Representativeness is deemed to have been achieved if the minimum sample size ( $n$ ) is less than or equal to the number of responses identifying a category as relevant. In accordance with Lohr (1999) [33],  $n$  is determined as shown in equation (2). In this regard,  $z$  represents the width of the confidence interval which in turn is based on the confidence level and can be taken from corresponding tables. For this study a confidence level of 95 % was assumed. Using this information,  $z$  is 1.96 which can be obtained by referring to Ref. [34].  $p_R$  represents the share of the characteristic "relevant" ( $C_R$ ) in  $N$  and can be calculated by using equation (3). The margin of error ( $E$ ) is set to 5 %.

$$n = \frac{z^2 * p_R * (1 - p_R)}{E^2} \quad (2)$$

$$p_R = \frac{C_R}{N} \quad (3)$$

In some cases, it is possible that the sample size  $n$  exceeds the total number of responses  $N$ . Therefore, the corrected sample size  $n_{corr}$  needs to be calculated as shown in equation (4).

$$n_{corr} = \frac{n}{\left(1 + \frac{n-1}{N}\right)} \quad (4)$$

Afterwards, the analysis of quantified GHG emissions is conducted on a category-specific basis for the identified corporate cluster. In this regard, the minimum, lower quartile, median, upper quartile, and maximum values are identified for each category, considering responses of zero emissions as well as those without. Based on this, the contribution of each category to the total amount of Scope 3 emissions within one corporate cluster is calculated referring to the median GHG emissions. To investigate if the results can be considered representative for the identified corporate cluster, an examination takes place based on equations (2)–(4). In accordance with the previous examination, the confidence level is set to 95 % and the margin of error ( $E$ ) is set to 5 %. The group under consideration is no longer the total number of responses ( $N$ ), but only those companies that have identified a category as "relevant" ( $N_R$ ). This assumption is based on the idea that the results are not meaningful for companies that have already identified a category as "not relevant". Moreover, the decisive characteristic affecting  $p$  is the number of companies that have identified a category as "relevant and calculated" ( $C_{R,c}$ ). The corresponding adaptations to equations (2)–(4) can be found in equations (5)–(7).

$$n_R = \frac{z^2 * p_{R,c} * (1 - p_{R,c})}{E^2} \quad (5)$$

$$p_{R,c} = \frac{C_{R,c}}{N_R} \quad (6)$$

$$n_{corr,R} = \frac{n_R}{\left(1 + \frac{n_R-1}{N_R}\right)} \quad (7)$$

## 4. Results

In this section, the results are presented starting with the variance and cluster analysis. Afterwards, the results of the benchmarking are illustrated at first referring to categories being identified as relevant. Second, quantifying GHG emissions and benchmarking them on that



basis is considered.

#### 4.1. Cluster analysis

For investigating whether industry, company size, and geographic location have significant influence on the emissions with regard to each Scope 3 category, a one-way variance analysis was carried out. The p-value was determined for each of these factors and for each category, separately. The results can be taken from Table 4.

Table 4 shows that the geographic location has no significant influence on the amount of emissions in any of the fifteen Scope 3 categories. With regard to the company size, this has only significant influence within categories 1, 10, and 12 to 14. Categories 13 and 14 are also significantly influenced by the industry. Based on these results and to ensure consistency across the following evaluations, corporate clusters are built considering company size and industry for all categories. According to equation (1), there is a potential for 28 clusters to emerge. As illustrated by Table 5, the described procedure for the cluster analysis results in 4359 companies being assigned to 26 different clusters. Table 5 provides a complete overview of all clusters and the number of companies assigned to each.

When examining Table 5 in detail, the following four clusters emerge as those with the most responses: services | BE with 753 companies, services | SME with 342 companies, manufacturing | BE with 882 companies, and manufacturing | SME with 610 companies. Since these four clusters provide the largest data basis per cluster, they will be considered as examples in the further course of this study. Moreover, a cluster consisting of all 4359 companies is considered as comparative figure in the further analysis to illustrate the influence of the cluster analysis on the benchmarking. In the following, this cluster is referred to as cluster a). Moreover, in the following services | BE is referred to as cluster b), services | SME is referred to as cluster c), manufacturing | BE cluster is referred to as cluster d), and manufacturing | SME is referred to as cluster e).

#### 4.2. Benchmarking

This section focuses on the results of the benchmarking process itself. First, the relevance of categories as assigned by the companies themselves is presented. Second, the emissions are quantified. Last, both relevance and quantification are evaluated regarding their representativeness for the whole corporate cluster.

First, the different response options for each Scope 3 category are examined regarding clusters a) to e). The results can be taken from Fig. 1. Fig. 1 depicts the subjective assessments of the companies regarding the relevance of each category to their activities.

Comparing Fig. 1b) and c) with Fig. 1d) and e) it becomes apparent that companies from the services industry tend to identify more categories as relevant than companies from the manufacturing industry. Moreover, manufacturing companies, with regard to the four considered clusters, seem to answer *Not evaluated* more often than services companies. Focusing on category 6 which represents business travel there are noticeable differences between the services clusters and the manufacturing clusters. Not only have more services companies identified this category as relevant but they have also calculated it more frequently. However, comparing BE and SME the individual identification of relevant categories is very similar. Nevertheless, regardless of industry and size, companies have identified downstream Scope 3 categories as *Not relevant* more often than upstream categories.

The subsequent analysis examines the representativeness of the initial findings using equations (2)–(4). It is determined whether a sufficient number of companies have identified a category as relevant or not relevant to allow this assumption to be representative for an entire cluster. Table 6 presents the results of this analysis. In general, it can be stated that the results regarding upstream categories are representative more often than those for downstream categories. The exception is

**Table 4**  
Overview of the results of the one-way variance analysis with points indicating where there is significant influence of a factor on the GHG emissions.

|              | Category 1 | Category 2 | Category 3 | Category 4 | Category 5 | Category 6 | Category 7 | Category 8 | Category 9 | Category 10 | Category 11 | Category 12 | Category 13 | Category 14 | Category 15 |
|--------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|
| company size | ●          |            |            |            |            |            |            |            |            | ●           |             | ●           |             | ●           |             |
| location     |            |            |            |            |            |            |            |            |            |             |             |             |             |             |             |
| industry     |            |            |            |            |            |            |            |            |            |             |             |             | ●           | ●           |             |

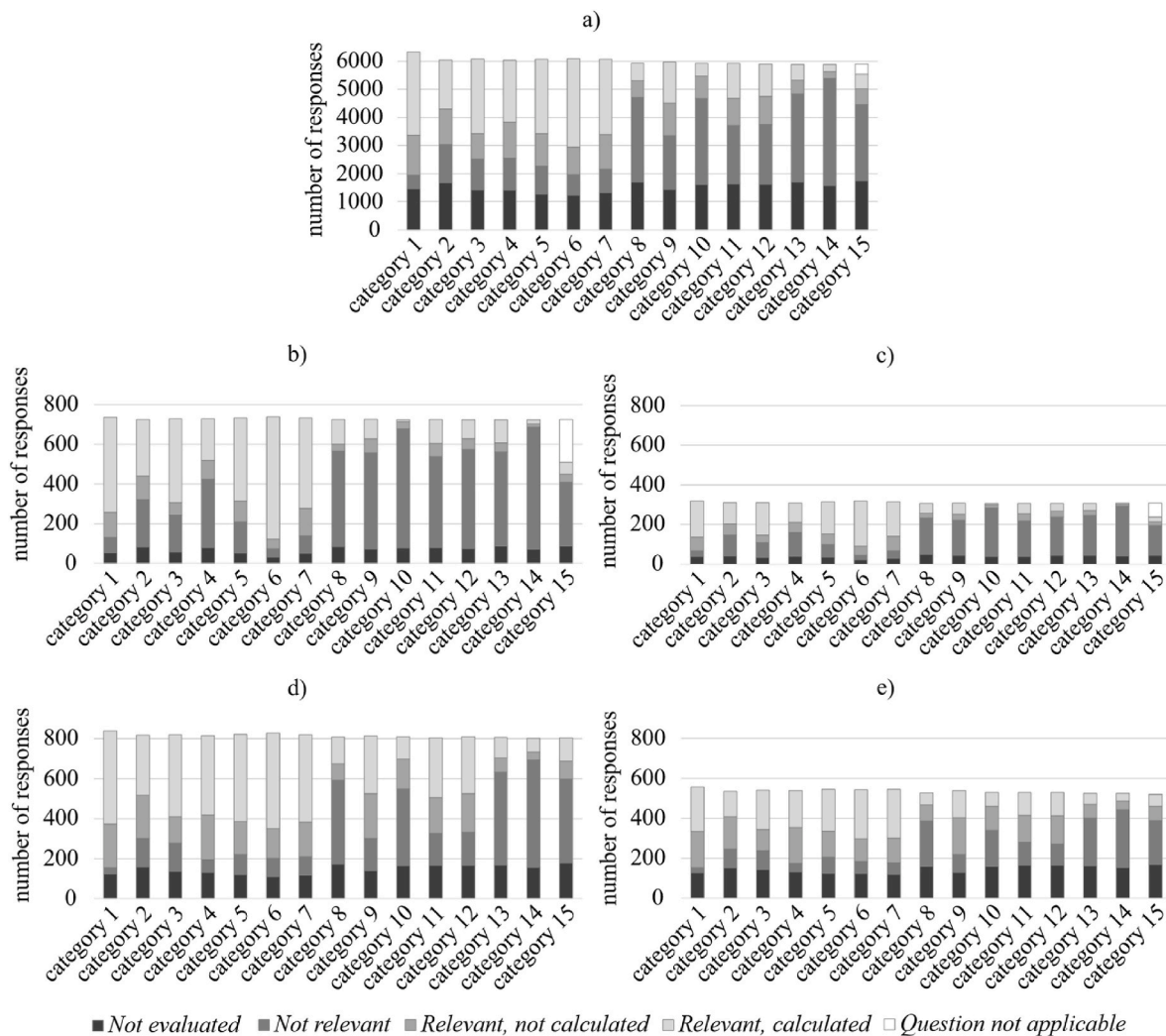
**Table 5**

Overview of the number of companies assigned to each cluster.

|                                 | BE  | SME |
|---------------------------------|-----|-----|
| services                        | 753 | 342 |
| infrastructure                  | 189 | 105 |
| biotech, health care and pharma | 98  | 50  |
| materials                       | 279 | 126 |
| manufacturing                   | 882 | 610 |
| hospitality                     | 32  | 17  |
| retail                          | 168 | 94  |
| transportation services         | 102 | 47  |
| food, beverage & agriculture    | 136 | 73  |
| power generation                | 53  | 19  |
| apparel                         | 62  | 33  |
| fossil fuels                    | 65  | 20  |
| corporate tags                  | 3   | 0   |
| international bodies            | 1   | 0   |

category 8 (upstream leased assets). For Purchased Goods and Services, Fuel- and Energy-Related Activities Not Included in Scope 1 or Scope 2, Waste Generated in Operations, Business Travel, and Employee Commuting, the results regarding the relevance of categories are representative for all four considered clusters. As one example, this means that other big manufacturing companies can assume that all upstream categories except for category 8 are relevant for them and should, therefore, be taken into account when evaluating Scope 3.

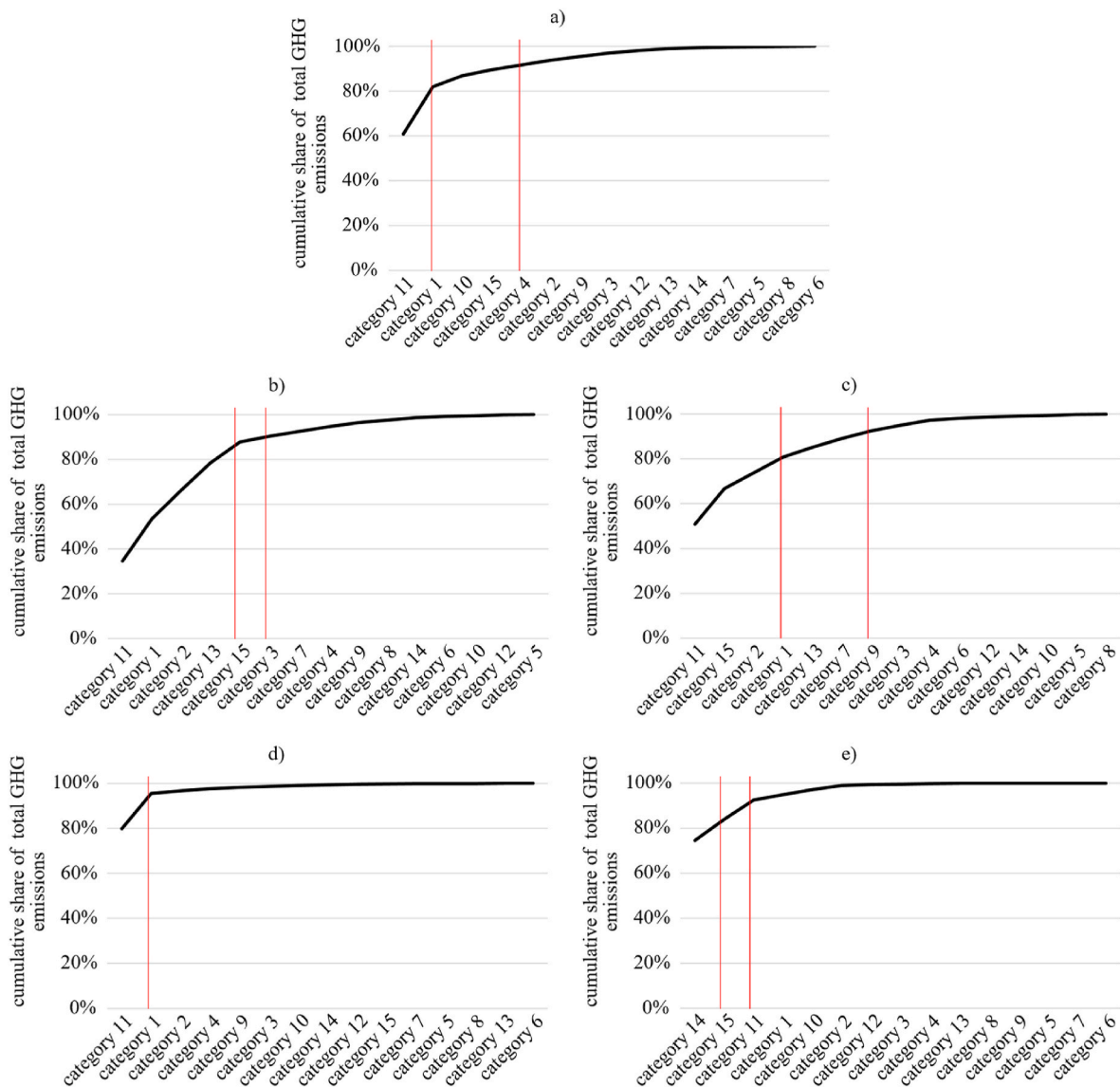
The quantification involves only those companies that have identified a category as relevant. Of particular interest are the responses from those companies that have already calculated the addressed categories. The cumulative share of each category in the total Scope 3 emissions is determined based on each category's median in t CO<sub>2</sub>e. The results are illustrated in Fig. 2 and exclude all responses that assign 0 t CO<sub>2</sub>e to a category. The same evaluation was also carried out for all responses including the assignment of 0 t CO<sub>2</sub>e. However, when determining the median values for each category, these resulted in 0 t CO<sub>2</sub>e especially for downstream categories. This can be attributed to too many companies declaring zero emissions to a category they had identified as relevant before. When analyzing which companies assigned relevant categories with 0 t CO<sub>2</sub>e it becomes evident that there are no significant differences between different, industries, locations, or company sizes. Rather, companies tend to allocate 0 t CO<sub>2</sub>e to downstream categories rather than to upstream categories. Nevertheless, assuming that if a category is valued at 0 t CO<sub>2</sub>e, it is either not relevant for the respective company or the corresponding GHG emissions have been set to net zero using off-setting certificates, this could lead to a misinterpretation of the actual results. Hence, this will not be further considered in this study. The first red line represents the border of 80 % and the second to 90 %. Both areas include the first category to cross the defined border to represent all those categories that make up at least 80 % or rather 90 % of the total GHG emissions in Scope 3.



**Fig. 1.** Proportional distribution of responses regarding a category's relevance referring to companies of cluster a) without clustering, b) services | BE, c) services | SME, d) manufacturing | BE, and e) manufacturing | SME.

**Table 6**  
Overview of representativeness of relevance and quantity of GHG emissions for each category referring to corporate clusters.

| category | services   BE |          | services   SME |          | manufacturing   BE |          | manufacturing   SME |          |
|----------|---------------|----------|----------------|----------|--------------------|----------|---------------------|----------|
|          | relevance     | quantity | relevance      | quantity | relevance          | quantity | relevance           | quantity |
| 1        | ●             | ●        | ●              | ●        | ●                  | ●        | ●                   |          |
| 2        | ●             | ●        |                | ●        | ●                  | ●        | ●                   |          |
| 3        | ●             | ●        | ●              | ●        | ●                  | ●        | ●                   |          |
| 4        | ●             | ●        |                | ●        | ●                  | ●        | ●                   |          |
| 5        | ●             | ●        | ●              | ●        | ●                  | ●        |                     | ●        |
| 6        | ●             | ●        | ●              | ●        | ●                  | ●        | ●                   | ●        |
| 7        | ●             | ●        | ●              | ●        | ●                  | ●        | ●                   | ●        |
| 8        |               | ●        |                |          |                    |          |                     |          |
| 9        |               |          |                |          | ●                  | ●        | ●                   |          |
| 10       |               |          |                |          | ●                  |          |                     |          |
| 11       |               |          |                |          | ●                  | ●        | ●                   |          |
| 12       |               |          |                |          | ●                  | ●        | ●                   |          |
| 13       |               | ●        |                |          |                    |          |                     |          |
| 14       |               |          |                |          |                    |          |                     |          |
| 15       |               |          |                |          |                    |          |                     |          |



**Fig. 2.** Cumulative share of Scope 3 categories referring to corporate clusters and based on median values excluding zero emissions.

When excluding all responses with 0 t CO<sub>2</sub>e, 80 % of Scope 3 emissions are made up by the Use of Sold Products (category 11), Purchased Goods and Services (category 1), Capital Goods (category 2), Downstream Leased Assets (category 13), and Investments (category 15) for BE of the services industry. Referring to SME of the same industry 80 % of Scope 3 emissions consist of categories 11, 15, 2, and 1. Thus, these two clusters show clear similarities within their emission hotspots. Focusing on Fig. 2d) which shows BE of the manufacturing industry it is categories 11 and 1 that only make up 80 %, but also 90 % of total Scope 3 emissions. This means, that companies of this cluster could focus on those two categories when assessing their Scope 3 and have potentially covered up most the GHG emissions of this Scope. When considering Fig. 2e), also category 11, the Use of Sold products, is of importance, but only when looking at the 90 % border. For the 80 % border, it is Franchises (category 14) and Investments (category 15) that clearly dominate Scope 3 emissions. Thus, companies of this cluster should focus on these two categories as their emission hotspots. However, referring to Fig. 2a) and, thus, to a benchmark without clustering at all, it is, again, categories 11 and 1 that generally dominate Scope 3. Comparing the four considered corporate clusters with this result it becomes apparent that these general dominating categories are of central importance even if considering corporate characteristics to increase the accuracy of the benchmarking. However, the clustering shows that next to these two categories also other categories such as Capital Goods, which is not even within the 90 % area for the general analysis, could be an emission hotspot depending on the individual industry.

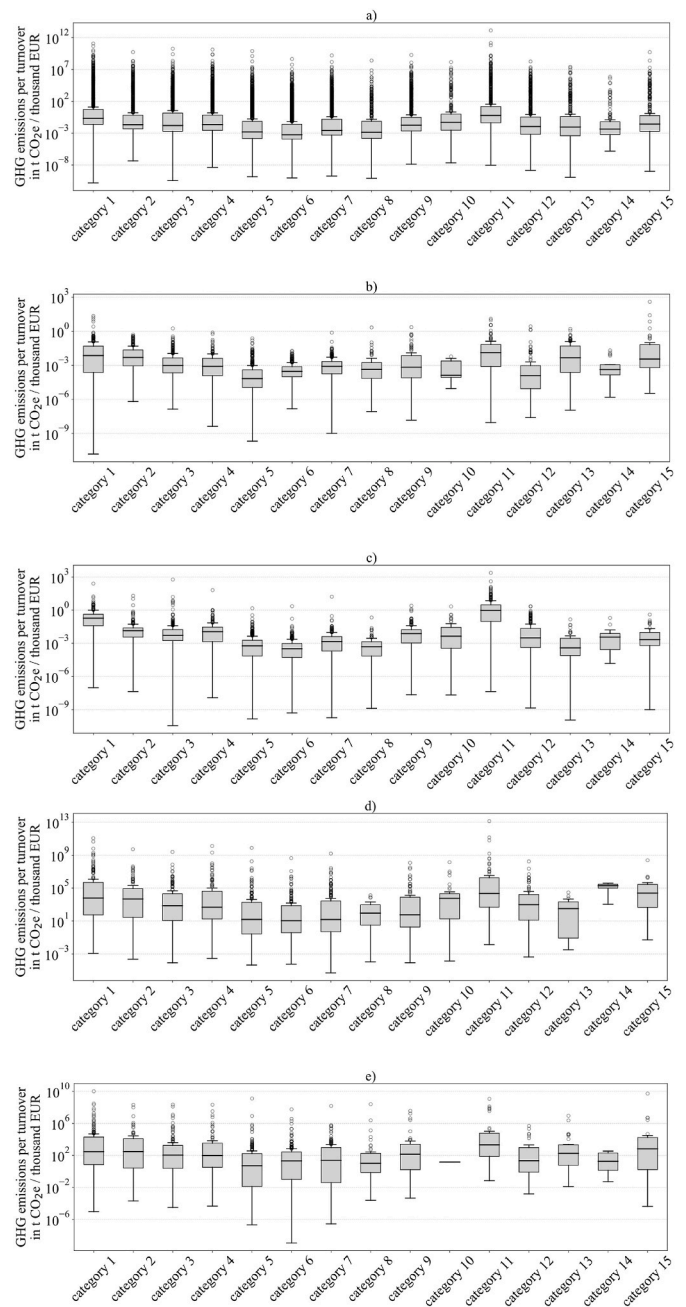
The analysis makes it evident that category 11 (Use of Sold Products) dominates Scope 3 emissions for companies in general with about 60 % on average. Moreover, downstream categories seem to be more relevant than upstream categories. However, depending on company size and industry, individual categories, such as category 2 (Capital Goods) for the services industry, can be identified as emission hotspots and, thus, be of central importance.

Afterwards, the second analysis of representativeness is conducted based on equations (5)–(7). The results indicate that across all four clusters within categories 5 (Waste Generated in Operations), 6 (Business Travel), and 7 (Employee Commuting) a sufficient number of companies quantified a category, which was identified as relevant before. Representativeness can be assumed for those. The cluster-specific results for all categories are presented in Table 6.

Table 6 provides an overview of the results of the two analyses for representativeness on a category-specific basis. The first criterion (relevance) indicates whether a sufficient number of companies have identified a category as relevant, allowing to be classified as representative for the entire cluster. The second criterion (quantity) indicates whether a sufficient number of companies have quantified the categories, making the corresponding results representative for those companies that have identified these categories as relevant. A point is used to indicate representativeness. If a category is representative with regard to both, relevance and quantity, this means that the quantification can not only be assumed to be representative for the companies that have identified the category as relevant, but for the entire cluster. In contrast, if there is only one checkmark in the quantity-column, the results are representative only for companies that have identified the category as relevant, but not for the entire cluster.

To conduct the benchmarking analysis, the absolute GHG emissions within each category were normalized with respect to the annual turnover, for obtaining a performance indicator expressed in t CO<sub>2</sub>e/thousand EUR. Fig. 3 illustrates the results without clustering and for clusters b) to e). In addition to the depiction of category-specific medians, the presentation encompasses visualizations of data uncertainties, particularly associated with outlier values. It must be mentioned that the scale on which the specific GHG emissions are indicated is shown logarithmically.

From Fig. 3, it becomes evident that all categories are associated with high uncertainties. Considering Fig. 3a) this mainly concerns category 1

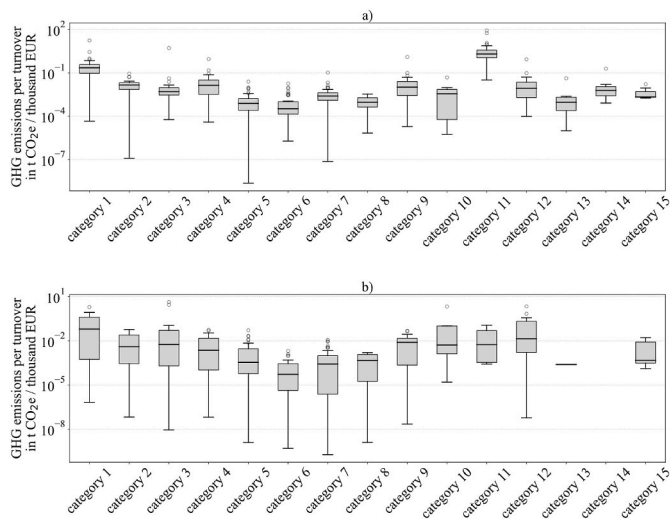


**Fig. 3.** Distribution of GHG emissions depending on the 15 Scope 3 categories for a) without clustering, b) services | BE, c) services | SME, d) manufacturing | BE, and e) manufacturing | SME, including the presentation of data uncertainties.

(Purchased Goods and Services) and category 11 (Use of Sold Products). Depending on company size and industry as well as amount of available data points different categories are accompanied by varying degrees of uncertainty.

Nevertheless, it must be taken into account that services, manufacturing and the other industries each comprise various sub-industries that may deviate from the overall results depicted in Fig. 3b)–3e). For this reason, cluster d) was further divided into subindustries. This cluster was chosen as it contains the largest number of datapoints with 882 companies. As examples, the subindustries automotive manufacturing and plastics manufacturing are chosen. The results are presented analogously to Fig. 3 and can be extracted from Fig. 4. Fig. 4a) illustrates automotive manufacturing and Fig. 4b) plastics





**Fig. 4.** Distribution of GHG emissions depending on the fifteen Scope 3 categories for a) automotive manufacturing and b) plastics manufacturing as examples of subindustries of cluster d) manufacturing | BE.

manufacturing.

For big enterprises in the subindustry automotive manufacturing it becomes apparent that composition and, hence, the emission hotspots in Scope 3 are very similar to superordinate cluster d) manufacturing | BE (Fig. 4a). For the subindustry plastics manufacturing the opposite is the case. Rather than by category 11, this subcluster is dominated by the Purchase of Goods and Services with about 60 % on average of total GHG emissions in Scope 3. The Use of Sold Products plays a clearly subordinate role at about 5 %.

## 5. Discussion

The results of the one-way variance analysis indicate that from the three potentially Scope 3 influencing factors – industry, company size, and geographic location – which were identified based on scientific literature, only two actually influence Scope 3 emissions. These two are the company size and the industry. This finding is in line with previous studies such as Bolay et al. [13] and Blanco et al. [10] that emphasize the influence of the industry on Scope 3. However, this study finds that both factors only have significant influence on certain categories, especially category 13 (Downstream Leased Assets) and category 14 (Franchises). Nevertheless, only highly aggregated industries were considered. When these are further divided into subindustries such as plastics or automotive manufacturing instead of just manufacturing it is conceivable that the industry could have significant influence on Scope 3 emissions regarding more categories. Against the background that other studies, such as those by Bolay et al. [13], have investigated Scope 3 but with a different division of industries, this appears interesting. Analogously, this could be the case for the geographic location. When dividing the defined six regions into smaller areas such as the M49 standard of the United Nations [35] does, these could influence Scope 3 emissions. Moreover, building clusters based on other criteria could lead to different results. An example could be the existence of a sustainability committee such as suggested by Córdova et al. [14]. However, such assumptions would have to be examined in further variance analyses. Moreover, further studies should consider varying the relevant variables to examine the impact on the results of both the cluster analysis and the benchmarking.

The results of the cluster analysis indicate that most companies are assigned to the industries manufacturing, services, materials, and infrastructure. Focusing on smaller clusters such as SME and BE in hospitality could reveal some more insights into such underrepresented

industries and it is conceivable that this could reveal clear differences between the conclusions drawn. However, a dataset consisting of more companies of those industries would be needed to be able to generate meaningful and representative results.

The findings indicate that Scope 3 emissions are dominated by category 11 (Use of Sold Products) with about 60 % on average. Regarding the considered clusters, category 11 is associated with a share of 9 % (manufacturing | SME) to 80 % (manufacturing | BE) of total Scope 3 emissions. When those two clusters that consider the industry services are taken into account, category 11 has a share of 30 % (BE) to 51 % (SME). This stays in contrast to the results of the study by Hansen et al., as this study identified category 1 (Purchased Goods and Services) in particular as the main driver of Scope 3 emissions [17]. Nevertheless, this category ranks second with a share of more than 20 % on average but ranging from 3 % (manufacturing | SME) to 19 % (services | BE) for the considered clusters. However, only the processing of food and beverages was explicitly considered in this study [17]. As Fig. 4 illustrates, the emission hotspots within the industry manufacturing can lie in different categories depending on the subindustry examined with downstream categories being of higher importance than upstream categories. However, Table 6 shows that very few downstream categories are representative in both relevance and quantity, meaning that not enough companies have provided information on these categories to make generalized statements for the respective clusters. This also applies to the Use of Sold Products (category 11). The results for this category are only fully representative for cluster d) (Table 6). The analysis also reveals that category 1, the Purchase of Goods and Services, dominates upstream emissions on average and also for three of the four considered clusters, which is in line with existing studies [5].

Referring to the results of the representativeness analyses, it was found that categories 5, 6, and 7 (Waste Generated in Operations, Business Travel, and Employee Commuting) are the only categories with full representativeness based on the underlying data. This may be attributed to good data availability regarding those categories. Nevertheless, when considering Fig. 2 it becomes evident that all three categories only exert a marginal influence on total Scope 3 emissions, especially compared to categories 1 and 11.

In total, the data is less representative for downstream activities compared to upstream activities (Table 6), which may be caused by an easier obtaining and evaluation of upstream data. This, in turn, could be the reason why no representative statements could be made for categories 14 and 15. Furthermore, accounting for GHG emissions stemming from these two categories as well as from category 8 (Upstream Leased Assets), category 10 (Processing of Sold Products), and category 13 (Downstream Leased Assets) may pose a special challenge for companies.

Regarding benchmarking, companies can compare their own activities in terms of corresponding emission intensity per yearly turnover with other companies in the same cluster allowing a category-specific corporate ranking. The results are based on highly aggregated characteristics. For instance, the industry manufacturing includes various sectors with different levels of emissions intensity – from steel production to the manufacturing of textiles. As the chosen examples – plastics and automotive manufacturing (Fig. 4) – illustrate, the sector-specific results may differ from the industry average. However, they can be used as a first approximation to estimate Scope 3 emissions of a category. Additionally, individual companies within each cluster may not fit the statements or conclusions made due to their unique circumstances. Therefore, it is essential to evaluate each company individually to determine whether they can be included in the benchmarking.

Furthermore, the data used in this study is based on voluntary information provided by the companies themselves and is not independently verified. This circumstance leads to inaccuracies in the reporting. It is stipulated that the total revenue, if chosen as a metric denominator, should be reported as a unit total in the CDP questionnaire. However, upon examining the revenues, it is observed that they are sometimes

provided with a shift by a factor of a thousand or even a million. When key performance indicators are subsequently derived based on these revenues, significant discrepancies can arise, particularly noticeable from Figs. 3 and 4. To minimize such inaccuracies, each company would need to be individually examined, to compare the reported data with corporate reporting that was verified by an independent third party. Such an investigation, along with potential implications for the formulation of key performance indicators, should be comprehensively scrutinized within additional studies. Nevertheless, it was assumed that the information provided by the companies is accurate throughout the study.

It is likely that further limitations come from various methodologies for calculating GHG emissions as this is not unified by the CDP questionnaire. Also, the selected emission factors, such as those for the procurement of goods, may vary even for the same product or material depending on the chosen database and, thus, impact the resulting GHG emissions. Moreover, as studies already indicate [17], it is likely that some companies did not consider all emission sources for each category, which further limits the comparability of the individual GHG emissions data across the different companies.

Due to the increasing reporting obligations for companies, it is assumed that the amount of reported data will increase, meaning that the analysis should be repeated in the coming years. However, companies criticize this increasing reporting obligations and are often overwhelmed by the complexity of Scope 3 reporting. This study has shown that there are hotspots in Scope 3 for every industry in certain categories. Using the CDP data and this analysis, a materiality analysis can be prepared for each industry and companies can be recommended to focus on the categories or to start analyzing emissions in this area. This reduces the overall scope, but at the same time identifies the main emissions and allows reduction measures with high savings potential to be developed.

## 6. Conclusions

In this study, Scope 3 emissions based on corporate clusters are investigated considering all 15 categories defined by the GHGP. Considering the results of a one-way variance analysis with which it was investigated whether industry, size, and location have significant influence on Scope 3 emissions, these clusters consider industry and company size. The results indicate that Scope 3 emissions are dominated by downstream categories, especially regarding the Use of Sold Products (category 11) with a share of total Scope 3 emissions ranging from 9 % to 80 %. When no clustering is considered at all, this category makes up more than 60 % of total Scope 3 emissions and is, thus, on average the biggest emission hotspot. Upstream categories are dominated by category 1 which represents the Purchase of Goods and Services. In dependence of the corporate cluster, this category has a share of total Scope 3 emissions ranging from 3 % to 19 % and being 20 % on average when no clusters are considered. This suggests that when assessing Scope 3 emissions, attention should be given to these two categories. However, depending on the cluster and whether further criteria are taken into account, also other categories can be emission hotspots.

Furthermore, it was shown that there are partially enormous deviations regarding the considered median values for all four corporate clusters investigated in this study. As some industries such as manufacturing comprise a variety of sectors from textiles to steel, subindustries were identified, and two examples presented for the cluster of big manufacturing enterprises. The considered subindustries – plastics and automotive – show that the results can vary from the industry average. Therefore, it is important to either cluster the companies with higher granularity or to assess each company individually to compare it only with those companies that have similar products or value chains.

The findings of this study have several implications for policymakers and regulatory bodies. The discrepancies in Scope 3 reporting practices across industries and company sizes highlight the need for standardized

reporting frameworks and guidelines. Policymakers should consider mandating Scope 3 reporting for large companies in emission-intensive sectors worldwide, similar to the Corporate Sustainability Reporting Directive in the European Union. Global efforts, such as the Greenhouse Gas Protocol, should be strengthened to promote consistent reporting standards. However, it should be taken into account whether individual industries may need to be considered and integrated separately, as they were underrepresented in this study, which only allows limited conclusions to be drawn in this regard.

Regarding implications for the companies themselves, identifying emission hotspots helps in implementing appropriate reduction measures. If categories 1 and 11 are designated as emission hotspots, a series of potentially suitable reduction measures emerges, particularly from the area of circular economy. For instance, purchasing recycled products or extending product usage phases could be possible measures. Referring to the Green Deal of the European Union, the introduction of a digital product passport is being discussed [36]. In addition to information on recycling rates, dismantlability and reparability, the digital product passport could also provide information regarding the GHG emissions [37], that are associated with a product or service (category 1). Such mandatory reporting could increase the comparability of the underlying data if standardized and verified by an independent third party. Moreover, within a digital product passport, a benchmarking analogous to that conducted in this study and performed explicitly for category 1 could be helpful for purchasing decisions. If a company decides in favor of a low-emission product this would reduce Scope 3, which, in turn, could possibly result in a competitive advantage in a society in which the reduction of GHG emissions is playing an increasingly important role. Moreover, companies can use the results of this study to estimate their Scope 3 emissions which is a necessary part of sustainability reporting, e. g., regarding the Corporate Sustainability Reporting Directive. Nevertheless, it should be taken into account that, depending on industry and size, other Scope 3 categories may represent emission hotspots. Therefore, the industries and sectors underrepresented in this study should be examined in further studies based on the same methodology. In addition to the CDP data, other databases could be taken into account for this to adequately consider regions in South America or Africa as well as industries such as power generation or apparel.

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## CRediT authorship contribution statement

**Nadja Buchenau:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Visualization, Supervision. **Jannik Oetzel:** Validation, Writing – original draft, Data curation, Formal analysis, Investigation, Visualization. **Ron-Hendrik Hechelmann:** Methodology, Investigation, Data curation, Visualization, Writing – original draft.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The authors do not have permission to share data due to legal reasons.

## References

- [1] Intergovernmental Panel on Climate Change, editor. Synthesis Report of the IPCC sixth assessment Report (AR6): longer Report; 2023 [Online]. Available: <https://www.ipcc.ch/report/ar6/syr/>. [Accessed 9 April 2023].
- [2] Ritchie H, Rosado P, Roser M. Emissions by sector. 2020 [Online]. Available: <https://ourworldindata.org/emissions-by-sector> [Accessed: 14 April 2023].
- [3] European Commission. Special eurobarometer 513 climate change. Fieldwork; March - April 2021, 2021 [Online]. Available: <https://europa.eu/eurobarometer/surveys/detail/2273>. [Accessed 14 April 2023].
- [4] World Business Council for Sustainable Development and World Resources Institute, editor. The greenhouse gas Protocol: a corporate accounting and reporting standard. Revised Edition; 2004 [Online]. Available: <https://ghgprotocol.org/corporate-standard>. [Accessed 6 March 2023].
- [5] Nill M, Schampel C, Schäpers R, Barsch A, Kurz K. Die Zukunft der globalen Wertschöpfung: Wettbewerbsfaktor Management der Scope-3-Emissionen der Lieferkette Analyse der 350 größten börsennotierten Unternehmen in der DACH-Region [Online]. Available: <https://sustain.com/einblicke/publikationen/>. [Accessed 14 April 2023].
- [6] Ducoulombier F. Understanding the importance of scope 3 emissions and the implications of data limitations. The Journal of Impact and ESG Investing 2021. <https://doi.org/10.3905/jesg.2021.1.018>. jesg.2021.1.018.
- [7] Schmidt M, Nill M, Scholz J. Determining the scope 3 emissions of companies. Chem Eng Technol 2022;45(7). <https://doi.org/10.1002/ceat.202200181>.
- [8] Klaaßen L, Stoll C. Harmonizing corporate carbon footprints. Nat Commun 2021; 12. <https://doi.org/10.1038/s41467-021-26349-x>.
- [9] Montoya-Torres JR, Gutierrez-Franco E, Blanco EE. Conceptual framework for measuring carbon footprint in supply chains. Prod Plann Control 2014. <https://doi.org/10.1080/09537287.2014.894215>.
- [10] Blanco C, Caro F, Corbett CJ. The state of supply chain carbon footprinting: analysis of CDP disclosures by US firms. J Clean Prod 2016;135:1189–97. <https://doi.org/10.1016/j.jclepro.2016.06.132>.
- [11] Rizet C, Browne M, Cornelis E, Leonardi J. Assessing carbon footprint and energy efficiency in competing supply chains: review – Case studies and benchmarking. Transport Res Part D 2012;17:293–300. <https://doi.org/10.1016/j.trd.2012.01.002>.
- [12] Acquaye A, Genovese A, Barrett J, Koh SCL. Benchmarking carbon emissions performance in supply chains. Supply Chain Manag: Int J 2014;19(3):306–21. <https://doi.org/10.1108/SCM-11-2013-0419>.
- [13] Bolay A-F, Bjørn A, Weber O, Margni M. Prospective sectoral GHG benchmarks based on corporate climate mitigation targets. J Clean Prod 2022;376:134220. <https://doi.org/10.1016/j.jclepro.2022.134220>.
- [14] Córdova C, Zorio-Grima A, Merello P. Carbon emissions by South American companies: Driving factors for reporting decisions and emissions reduction. Sustainability 2018;10. <https://doi.org/10.3390/su10072411>.
- [15] Schulman DJ, Bateman AH, Greene S. Supply chains (Scope 3) toward sustainable food systems: an analysis of food & beverage processing corporate greenhouse gas emissions disclosure. Cleaner Production Letters 2021;1. <https://doi.org/10.1016/j.clpl.2021.100002>.
- [16] Shirmali G. Scope 3 emissions: measurement and management. The Journal of Impact and ESG Investing 2022;3(1):31–54. <https://doi.org/10.3905/jesg.2022.1.051>.
- [17] Hansen AD, Kuramochi T, Wicke B. The status of corporate greenhouse gas emissions reporting in the food sector: an evaluation of food and beverage manufacturers. J Clean Prod 2022;361:132279. <https://doi.org/10.1016/j.jclepro.2022.132279>.
- [18] Asif MS, Lau H, Nakandala D, Fan Y, Hurriyet H. Case study research of green life cycle model for the evaluation and reduction of scope 3 emissions in food supply chains. Corp Soc Responsib Environ Manag 2022;29:1050–66.
- [19] Radonjić G, Tompa S. Carbon footprint calculation in telecommunications companies – the importance and relevance of scope 3 greenhouse gases emissions. Renew Sustain Energy Rev 2018;98:361–75. <https://doi.org/10.1016/j.rser.2018.09.018>.
- [20] Budolfson MB, et al. Utilitarian benchmarks for emissions and pledges promote equity, climate and development. Nat Clim Change 2021;11:827–33. <https://doi.org/10.1038/s41558-021-01130-6>.
- [21] Guevara M, Lopez-Aparicio S, Cuvelier C, Tarrason L, Clappier A, Thunis P. A benchmarking tool to screen and compare bottom-up and top-down atmospheric emission inventories. Air Quality, Atmosphere & Health 2017;10(5):627–42. <https://doi.org/10.1007/s11869-016-0456-6>.
- [22] Jeong K, Hong T, Kim J. Development of a CO2 emission benchmark for achieving the national CO2 emission reduction target by 2030. Energy Build 2018;158: 86–94. <https://doi.org/10.1016/j.enbuild.2017.10.015>.
- [23] Zipperer V, Sato M, Neuhoft K. Benchmark for emissions trading - general principles for emissions scope: discussion papers. 1712. Berlin: Deutsches Institut für Wirtschaftsforschung; 2017. p. 3–7 [Online]. Available: [https://www.diw.de/de/diw\\_01.c.574085.de/publikationen/diskussionspapiere/2017\\_1712/benchmarks\\_for\\_emissions\\_trading\\_\\_\\_general\\_principles\\_for\\_emissions\\_scope.html](https://www.diw.de/de/diw_01.c.574085.de/publikationen/diskussionspapiere/2017_1712/benchmarks_for_emissions_trading___general_principles_for_emissions_scope.html) [Accessed 28 April 2023].
- [24] Mejia C, Kajikawa Y. Estimating scope 3 greenhouse gas emissions through the shareholder network of publicly traded firms. Sustain Sci 2024;19(4):1409–25. <https://doi.org/10.1007/s11625-023-01460-8>.
- [25] World Resources Institute and World Business Council for Sustainable Development, editor. Technical Guidance for calculating scope 3 emissions: Supplement to the corporate value chain (scope 3) Accounting & reporting standard; 2013 [Online]. Available: version 1.0. <https://ghgprotocol.org/scope-3-technical-calculation-guidance>. [Accessed 6 March 2023].
- [26] Backhaus K, Erichson B, Gensler S, Weiber R, Weiber T. *Multivariate Analysemethoden: Eine anwendungsorientierte Einführung*. 16. vollständig überarbeitete und erweiterte Auflage. Springer Gabler; 2021.
- [27] Stocker TC, Steinke I. Statistik: Grundlagen und Methodik. Berlin, Boston: De Gruyter Oldenbourg; 2017.
- [28] Chandrakantha L. Learning ANOVA concepts using simulation. In: Proceedings of the 2014 Zone 1 Conference of the American society for Engineering Education; 2014. <https://doi.org/10.1109/ASEEZone1.2014.6820644>.
- [29] Madhulatha TS. An overview on clustering methods. International Organization of Scientific Research Journal of Engineering 2012;4(4):719–25. <https://doi.org/10.48550/arXiv.1205.1117>.
- [30] Huang Z. Extensions to the k-means algorithm for clustering large data sets with categorical values. Data Min Knowl Discov 1998;2(2):283–304. <https://doi.org/10.1023/A:1009769707641>.
- [31] European Commission. User guide to the SME definition [Online]. Available: <https://ec.europa.eu/docsroom/documents/42921>. [Accessed 18 April 2023].
- [32] Optiz O, editor. Conceptual and Numerical analysis of data. Berlin, Heidelberg: Springer Berlin Heidelberg; 1989.
- [33] Lohr SL. Sampling: Design and analysis. Pacific Grove, Albany, Belmont, Bonn, Boston, Cincinnati, Detroit, Johannesburg, London, Madrid, Melbourne, Mexico City, New York, Paris, Singapore, Tokyo, Toronto, Washington. Duxbury Press; 1999.
- [34] Papula L. Mathematik für Ingenieure und Naturwissenschaftler Band 3: Vektoranalysis, Wahrscheinlichkeitsrechnung, Mathematische Statistik, Fehler- und Ausgleichsrechnung. Springer Fachmedien Wiesbaden; 2016.
- [35] United Nations, Methodology: Standard country or area codes for statistical use (M49). [Online]. Available: <https://unstats.un.org/unsd/methodology/m49/> (accessed: May 17 2024).
- [36] European Commission, Green Deal: New proposals to make sustainable products the norm and boost Europe's resource independence. [Online]. Available: [https://ec.europa.eu/commission/presscorner/detail/en/ip\\_22\\_2013](https://ec.europa.eu/commission/presscorner/detail/en/ip_22_2013) (accessed: April 27 2023).
- [37] World Business Council for Sustainable Development and Boston Consulting Group. Enabling circularity through transparency: introducing the EU digital product passport [Online]. Available: <https://www.wbcsd.org/Pathways/Products-and-Materials/Resources/The-EU-Digital-Product-Passport>. [Accessed 27 April 2023].